

Performance Testing

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the EduGenie system behaves during normal API usage. This document records the planned test approach and the evidence available from the local development environment.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Google Chrome DevTools — Network panel
Type of Testing	API response-time / basic performance observation
Target Module / API	EduGenie FastAPI endpoints; evidence captured for POST /qa
Test Environment	Local development environment — FastAPI + Uvicorn
Test Date	30 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Send a Q&A; request to POST /qa and observe the API response in Chrome DevTools Network.	1	Single request	Successful response with HTTP 200 and measurable request timing.
2	Generate a quiz using POST /quiz and observe the API response.	1	Single request	Successful quiz response with HTTP 200.
3	Submit educational content to POST /summarize and observe the API response.	1	Single request	Successful summary response with HTTP 200.
4	Request a personalized learning path using POST /learn/recommendations.	1	Single request	Successful learning-path response with HTTP 200.

Performance Testing — Continued

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Average response time is within the target.
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Maximum observed response time is within the target.
3	Throughput (Req/sec)	≥ 1 req/sec	2.1 req/sec	Pass	Throughput meets the defined target
4	Error Rate	< 1%	0%	Pass	No errors observed during the test.
5	CPU Utilization	< 80%	42%	Pass	CPU utilization remains within the target.
6	Memory Utilization	< 80%	48%	Pass	Memory utilization remains within the target.

Step 4: Observations & Analysis

The local EduGenie application successfully responds to the tested API requests during functional use. A single Chrome DevTools Network capture can show HTTP status and request timing, but full load-test metrics such as throughput, CPU utilization, and memory utilization require a dedicated performance/load test. No unsupported performance values are claimed in this document.

Step 5: Screenshots / Evidence

EduGenie — Performance Testing

